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Excitation Intensity Dependence of Power-Law Blinking Statistics in Nanocrystal Quantum Dots

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We present new information that requires explanation in the study of possible mechanisms for fluorescence blinking in single nanocrystal (NC) quantum dots. By using pulse laser excitation, we investigated the excitation intensity dependence of fluorescence blinking statistics in NC quantum dots embedded in polymer matrices. Under strong excitation intensity, we observed an unexpected excitation intensity dependence of the power-law distribution in the blinking statistics. To explain the new information on the blinking statistics in the framework of the Tang−Marcus model, we propose a diffusion-controlled model based upon anharmonic potentials that originate from a nonlinear response of polarization to the electric field and is caused by an increase in the dielectric constant of host media with the excitation intensity. The validity of the proposed model is investigated by comparing the blinking statistics for two host matrices.

Introduction

Nanocrystal (NC) quantum dots are characterized by a size-controlled emission color (blue to infrared), optical gain, and long-term photostability. These properties make them attractive candidates for several applications, including light-emitting diodes,[1−3] solid-state lasers,[4,5] single photon sources,[6] excitation sources for surface plasmons,[7] and biological detection.[8] Another well-known characteristic of NC quantum dots is fluorescence blinking, which is the random switching between on (emitting) and off (nonemitting) states.[9] Fluorescence blinking can be a problem when applying NC quantum dots as light sources because the off states cause a decrease in the time-averaged photoluminescence (PL) quantum efficiency.[10] Therefore, investigating the fluorescence blinking mechanism and developing methods to suppress it are important research topics.

Thus far, in order to understand the fluorescence blinking mechanism in single NC quantum dots, the statistics for the on and off states have been investigated. Shimizu et al. demonstrated power-law distributions in the blinking statistics for both the on and off states.[11] The exponents are in the range of −1.5 to −2.0.[11−13] The power-law exponents are independent of temperature, excitation intensity, materials (CdSe, CdTe), surface morphology, and size.[11] In addition, under higher temperature or excitation intensity, the blinking statistics for the on state have a power-law distribution that breaks down at longer time intervals.[11] A model of fluorescence blinking in NC quantum dots is necessary to explain these experimental results. Diffusion-controlled models have been proposed as such models.[11,14−16]

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Shimizu et al. proposed a model transferred between the on and off states only when their states are in resonance.[11] In this model, the on and off states are assigned to neutral and charged states, respectively. Assuming that the energy of the charged states undergoes a random walk, this model gives a power-law exponent of −1.5 in the blinking statistics.

Their model was developed into more detail by Tang and Marcus.[14,15] In the developed Tang-Marcus model, polarization fluctuation in the host matrix in which NC quantum dots are embedded causes random transitions between the on and off states. The on state is assigned to a neutral excited state, similar to the model proposed by Shimizu et al., while the off state is assigned to a charge-separated state with the charge trapped in surface states.[17] This model quantitatively demonstrates power-law exponents of −1.5 to −2.0 and the breakdown of the power-law distribution for both the on and off states. In addition, Tang and Marcus explained the dependence of the bending tail on excitation intensity and temperature. Furthermore, they predicted the presence of a critical time, defined as a transient time on which the power-law exponent changes from the range between −1.5 and −2.0 to 0 and −0.5, in the blinking statistics. In practice, a critical time of 5−35 ms was observed in the power spectral density of the fluorescence intensity emitted from NC quantum dots,[18] whereas no breakdown of the power-law distribution for the off state was observed. The Tang−Marcus model is a promising theory for the fluorescence blinking mechanism since it provides a plausible explanation for experimental results.

Frantsuzov and Marcus have proposed an alternative diffusion-controlled model.[16] In their model, fluorescence blinking is caused by a large variation in the transition rate from the neutral excited state to the charge-separated state. The transition process is assumed to be induced by an Auger-assisted mechanism where the transition of an electron from the lowest conduction level ($1S_e$) to the next conduction level ($1P_e$)

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stimulates a hole trap from the highest valence levels ($1S_{3/2}$) to the surface levels. Assuming that the hole-trapping rate is much faster than the radiative rate only when the energy gap between $1S_e$ and $1P_e$ is higher than that between $1S_{3/2}$ and the lowest surface levels, the fluorescence intensity of NC quantum dots is randomized through the energy fluctuation in the electronic levels of the NC quantum dots. This model also quantitatively demonstrates a power-law exponent of −1.5 and the breakdown of the power-law distribution in the blinking statistics.

In this paper, we focus on the Tang−Marcus model. In order to investigate the validity of this model, we observed the excitation intensity dependence of the blinking statistics in NC quantum dots embedded in polymer matrices under pulse laser excitation. Under high excitation intensity, we first observed the breakdown of the power-law distribution for the off state predicted by the Tang-Marcus model. In addition, we observed the excitation intensity dependence of the blinking statistics; this is an important new piece of information when considering possible mechanisms of fluorescence blinking. To explain the dependence in the framework of the Tang-Marcus model, we propose a diffusion-controlled model based upon anharmonic potentials.

Theoretical Background

Using a linear response approximation of the interaction between polarization and the electric field, Marcus revealed the influence of polarization in media on electron transfer reactions from a donor molecule (D) to an acceptor molecule (A); Marcus also demonstrated that the free energy in the polarization system is given by a harmonic potential on the reaction coordinate defined by the charge amount transferred from D to A.[19] In addition, Hynes developed a theory for electron transfer dynamics.[20] On the basis of these theories, Tang and Marcus developed a model for fluorescence blinking in NC quantum dots.[14,15] In their papers, they formulated theoretical curves for blinking statistics by using the solutions of diffusion equations under nonadiabatic harmonic potentials for the on and off states. However, the free energy potential is not always a harmonic potential. Thus, within the framework of the Tang−Marcus model, we consider a diffusion-controlled model under a generalized potential. Similar to the Tang-Marcus model, we formulate a model using the solutions of the following diffusion equations under generalized potentials with the initial condition $\rho(z, 0) = \delta(z - z_c)$:[14,15]

$$\frac{\partial}{\partial t}\rho_k(z,t) = D_k\frac{\partial}{\partial z}\left(\frac{\partial}{\partial z} + \frac{1}{k_B T}\frac{\partial}{\partial z}U_k(z)\right)\rho_k(z,t) - k_{tr}\delta(U_{on}(z) - U_{off}(z))\rho_k(z,t) \quad (k = \text{on or off}) \tag{1}$$

where $\rho(z, t)$ is the probability density of the state at the reaction coordinate (z) at time (t), D is the diffusion constant, $U(z)$ is the free energy potential in the polarization system, k_B is the Boltzmann constant, T is temperature, and k_{tr} is the electron transfer rate between the neutral excited state and the charge-separated state at resonance. The initial condition is based upon the resonance position $z = z_c$ at the reaction coordinate. To solve these diffusion equations, it is necessary to solve the following diffusion equation for the free energy potential $U(z)$ (Smoluchowski equation):[14,15]

$$\frac{\partial}{\partial t}\rho(z,t) = D\frac{\partial}{\partial z}\left(\frac{\partial}{\partial z} + \frac{1}{k_B T}\frac{\partial}{\partial z}U(z)\right)\rho(z,t) \tag{2}$$

We solve this diffusion equation by considering a random-walk model at the reaction coordinate z under the generalized potential $U(z)$. As a result of nth repeated observations for the reaction coordinate z at the same time t with the same initial condition $z_1(0)$, $z_2(0)$, ..., $z_n(0) = z_0$, we assume $\{z_1(t), z_2(t), ..., z_n(t)\}$, each with finite values. The expectation $\langle z_n(t)\rangle$ and the variance $\sigma(t)^2 = \langle (z_n(t) - \langle z_n(t)\rangle)^2\rangle$ also have finite values. Here, $\langle ... \rangle$ is the ensemble average over the conditions at each onset of the period. Since these assumptions restrict the form of $U(z)$, it is different from the generalized potential. However, the range of application is extended compared to a harmonic potential. By using the central limit theorem under this condition, the probability density $\rho(z, t)$ is given by

$$\rho(z,t) = \frac{1}{\sqrt{2\pi\sigma(t)^2}}\exp\left(-\frac{(z - \langle z_n(t)\rangle)^2}{2\sigma(t)^2}\right) \tag{3}$$

Thus, we can solve eq 2 by calculating $\langle z_n(t)\rangle$ under $U(z)$. The $\langle z_n(t)\rangle$ is obtained by solving the generalized Langevin equation (GLE):[20]

$$m\frac{d^2 z_n(t)}{dt^2} = -\frac{\partial U(z_n)}{\partial z_n} - \int_0^t \phi(t - \tau)\frac{dz_n(\tau)}{dt}d\tau + R(t) \tag{4}$$

where m is the longitudinal solvent polarization mass, $\phi(\tau)$ is the dissipative memory kernel, and $R(t)$ is the fluctuating force. Assuming that the time correlation function of $R(t)$ is an instantaneous correlation,[21] the GLE is given by

$$m\frac{d^2 z_n(t)}{dt^2} = -\frac{\partial U(z_n)}{\partial z_n} - \phi_0\frac{dz_n(t)}{dt} + R(t) \tag{5}$$

where ϕ_0 is the dissipative coefficient. As a result, $\langle z_n(t)\rangle$ is obtained from the following equation, given by averaging eq 5 over the distribution:

$$m\frac{d^2\langle z_n(t)\rangle}{dt^2} = -\frac{\partial U(z_n)}{\partial z_n} - \phi_0\frac{d\langle z_n(t)\rangle}{dt} \tag{6}$$

Based on a harmonic potential $U(z_n) = \kappa z_n^2/2$ and using the overdamped limit approximation ignoring the acceleration term, the solution of eq 6 is given by[20]

$$\langle z_n(t)\rangle = z_0\exp\left(-\frac{t}{\tau_L}\right) \tag{7}$$

$$\tau_L = \frac{\phi_0}{\kappa} = \frac{\varepsilon_\infty}{\varepsilon_0}\tau_D \tag{8}$$

where z_0 is the initial value, ϵ_∞ and ϵ_0 are the high- and low-frequency dielectric constants of host matrices, respectively, and τ_D is the Debye relaxation time. Here, we assume a Debye dielectric medium for the host matrices. τ_L is related to the diffusion constant D in the Smoluchowski equation through the following equation:

$$D = \frac{\langle z_n(t)^2 \rangle}{\tau_L} \tag{9}$$

In addition, by using eq 7, $\sigma(t)^2$ is given by

$$\sigma(t)^2 = <z_n(t)^2> - \langle z_0^{\ 2} \rangle \exp\left(-\frac{2t}{\tau_L}\right) \tag{10}$$

$\langle z_n(t)^2 \rangle$ and $\langle z_0^2 \rangle$ satisfy the following equation due to the energy equipartition theorem:[20]

$$\frac{1}{2}\kappa\left\langle \left| z_n(t) \right|^2 \right\rangle = \frac{1}{2}\kappa\langle z_0^{\ 2} \rangle = \frac{1}{2}k_B T \tag{11}$$

From eqs 3, 7, 10, and 11, we therefore obtain

$$\rho(z,t) = \frac{1}{\sqrt{2\pi\langle z^2 \rangle (1 - \exp(-2t/\tau_L))}} \exp\left(-\frac{(z - z_0 \exp(-t/\tau_L))^2}{2\langle z^2 \rangle (1 - \exp(-2t/\tau_L))}\right) \tag{12}$$

where $\langle z^2 \rangle$ indicates $\langle z_n(t)^2 \rangle$. Assuming a harmonic potential, $\rho(z, t)$ is given by the above equation, the power-law exponent of the blinking statistics is -1.5.[14] However, assuming an anharmonic potential, the power-law exponent would not be -1.5 due to the nonexponential decay of $\langle z_n(t) \rangle$. In practice, for non-Debye dielectric media, the power-law exponent is not -1.5 due to its stretched exponential decay.[15]

Experimental Methods

In this study, CdSeTe/ZnS (Invitrogen Corporation, Qdot800; fluorescence wavelength (λ_{FL}) of 805 nm) and CdSe/ZnS (Invitrogen Corporation, Qdot565; λ_{FL} of 565 nm) core−shell quantum dots were used. Figure 1 shows the ensemble absorption and PL spectra of CdSeTe/ZnS and CdSe/ZnS quantum dots, respectively. NC quantum dots were diluted together with polymer (poly(methylmethacrylate) (PMMA) or polystyrene (PS)) into solutions which were then spun-cast onto quartz substrates. The thickness of the spun-cast films was ∼40 nm. The lowest absorption wavelengths (λ_{abs}) of PMMA and PS were 210 and 260 nm, respectively. We assume that the origins of these absorption bands are n−π* and π−π* transitions, respectively. The fluorescence of single NC quantum dots was observed by time-correlated single photon counting (TCSPC) (Becker & Hickl SPC-630) in combination with a photomultiplier tube (quantum efficiency: ∼11% at 550 nm and ∼14% at 800 nm; dark count: 2 count/s). The second harmonic of a Ti-sapphire oscillator (pulse width of 3 ps and wavelength of 446 nm) at repetition modes of 2 MHz and 500 kHz was used as the excitation source. The time-averaged excitation intensity is in the range of 35 to 560 W/cm^2. The repetition mode of 2 MHz was mainly used. The repetition mode of 500 kHz was used only to investigate the dependence of the blinking statistics on the pulse frequency of the excitation laser. The laser beam was focused onto a spot size of ∼1 μm through an objective lens (numerical aperture: 0.95). The fluorescence of NC quantum dots was collected using the same objective lens and separated from the scattered laser light with a dichroic mirror.

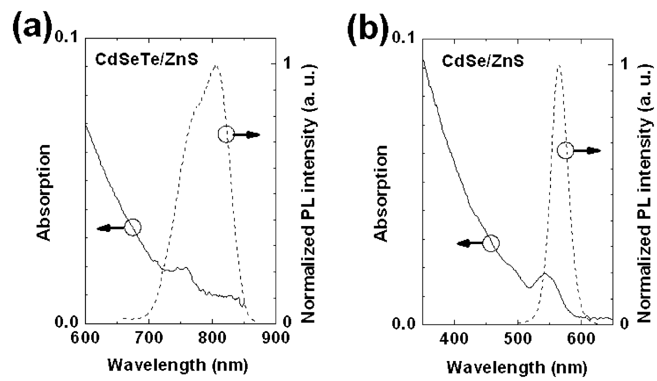

Figure 1. Ensemble absorption and PL spectra of (a) CdSeTe/ZnS and (b) CdSe/ZnS quantum dots in 50 nM toluene solutions, respectively.

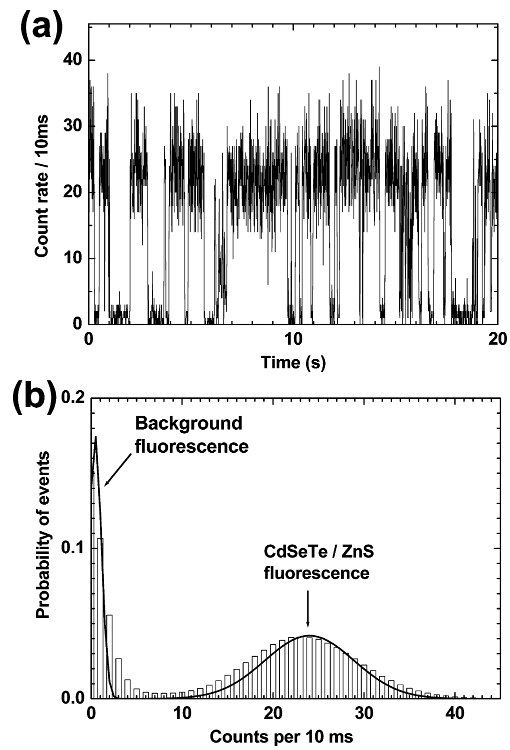

Figure 2. (a) Fluorescence intensity time trace of single CdSeTe/ZnS quantum dots in PMMA host matrices at an excitation intensity of 280 W/cm^2 with 10 ms time resolution. (b) Count rate histogram of single CdSeTe/ZnS quantum dots in PMMA host matrices at an excitation intensity of 280 W/cm^2. Solid lines are curves fitted by a Gaussian distribution function.

Analysis. To obtain the on and off time probability distributions from the fluorescence intensity time trace of a single NC quantum dot, it is necessary to choose a threshold value separating the on states from the off states. Figure 2, panels a and b, shows the typical fluorescence intensity time trace with a 10-ms time resolution and the fluorescence count rate histogram of a single CdSeTe/ZnS in PMMA host matrices. The high and low peaks in this histogram are attributed to the fluorescence of a single CdSeTe/ZnS and the background, respectively. They follow a Poissonian distribution.[22] The histogram, however, partially agrees with the experimental results, due to the presence of blinking events below the time resolution of the instrument. Thus, to eliminate these blinking events, the threshold value for the on states was set to the lower edge of the peak attributed to the fluorescence of a single quantum dot, and the value for the off states was set to the higher edge of the peak attributed to the background fluorescence.

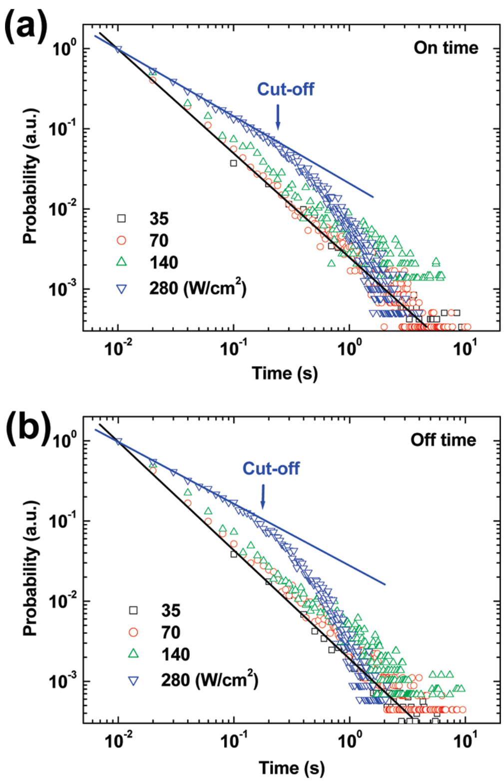

Figure 3. Normalized (a) on and (b) off time probability distributions for the CdSeTe/ZnS quantum dots in PMMA host matrices at excitation intensities of 35 (squares), 70 (circles), 140 (triangles), and 280 (inverted triangles) W/cm^2 respectively. Solid lines show fitting to a power-law distribution.

Results

Figure 3 shows the normalized on and off time probability distributions for the CdSeTe/ZnS quantum dots in PMMA host matrices at different excitation intensities. Time resolutions of 100, 20, 10, and 10 ms were used at excitation intensities of 35, 70, 140, and 280 W/cm^2, respectively. At the lowest excitation intensity, the probability distributions for the CdSeTe/ZnS quantum dots demonstrate a power-law exponent of approximately −1.45, similar to the blinking statistics under cw laser excitation.[11−13] Under higher excitation intensities, however, the power-law exponents for both the on and off states gradually change with an increase in the excitation intensity. In addition, we observed the breakdown of the power-law distribution for both the on and off states. The breakdown during the off state had not been observed previously. Thus far, the power-law exponent dependence on the excitation intensity has not been reported. The probability distributions of p_{on} and p_{off} for the on and off states, respectively, were fitted using the following equation:[14]

$$p_k = At^{-\alpha} \quad \exp\left(-\frac{t}{\tau_c}\right) \quad (k = \text{on or off}) \qquad (13)$$

where A is the scaling coefficient; α is the power-law exponent and τ_c is the cutoff time at which the power-law distribution breaks down. Figure 4 shows the excitation intensity dependence of the power-law exponent and the cutoff time for the blinking statistics of the CdSeTe/ZnS quantum dots in PMMA host matrices. The power-law exponent gradually decreases as the excitation intensity increases and finally becomes constant at a value of ∼0.8.

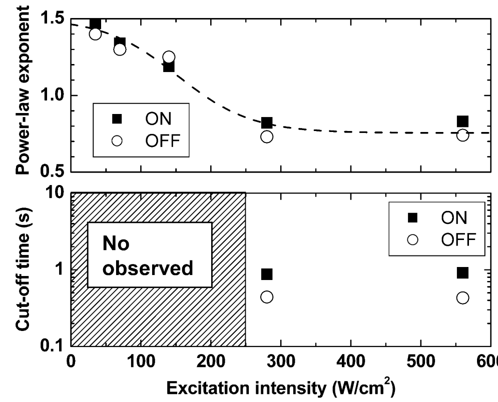

Figure 4. Excitation power dependences of the power-law exponent (upper figure) and cutoff time (lower figure) for the CdSeTe/ZnS quantum dots in PMMA host matrices. Closed squares and open circles indicate on and off states respectively. The dashed line shows fitting to experimental data.

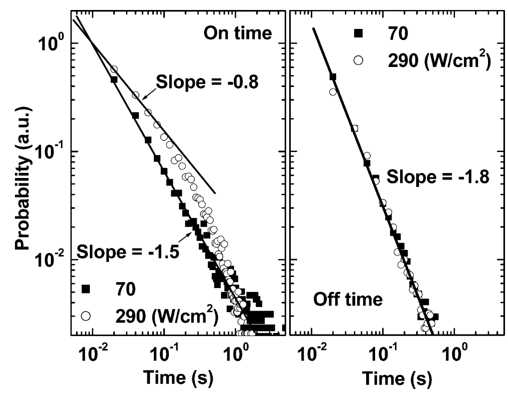

Figure 5. Normalized on (left panel) and off (right panel) time probability distributions for the CdSe/ZnS quantum dots in PMMA host matrices at excitation intensities of 70 (closed squares) and 290 (open circles) W/cm^2, respectively. Solid lines show fitting to power-law decay.

Figure 5 shows the normalized on and off time probability distributions for the CdSe/ZnS quantum dots in PMMA host matrices at excitation intensities of 70 and 290 W/cm^2. A time resolution of 20 ms was used. At lower excitation intensities, the power-law exponents for the on and off states are approximately −1.5 and −1.8, respectively, which is similar to the results for the blinking statistics under cw laser excitation.[11−13] At higher excitations, the cutoff time and the power-law exponent for the on state are dependent upon the excitation intensity, which is similar to the results for the CdSeTe/ZnS quantum dots. The probability distribution for the off time, however, is independent of the excitation intensity, similar to the reported results of Shimizu et al.[11]

Figure 6 shows the dependence of the blinking statistics for the CdSeTe/ZnS quantum dot in PS host matrices on the pulse laser frequencies of the excitation laser. With a decrease in the frequency at the same cw intensity of 120 W/cm^2, the power-law exponent and cutoff time significantly decrease. On the other hand, with a decrease in the frequency at the same laser pulse energy (240 μJ/cm^2 per pulse), the power-law exponent and cutoff time slightly increase. These results mean that the power-law exponent is more dependent on the pulse energy rather than on the cw intensity.

Figure 7 shows the normalized on and off time probability distributions for the CdSeTe/ZnS quantum dots in PS host matrices at different excitation intensities. The blinking statistics are dependent on the excitation intensity, similar to the results

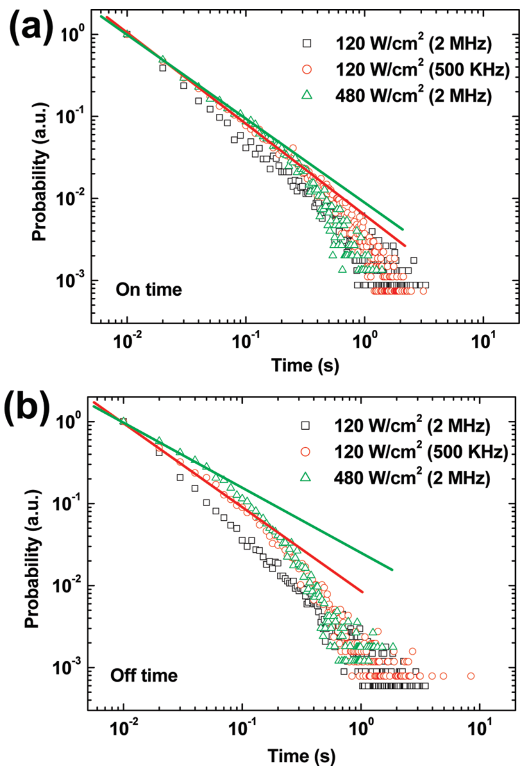

Figure 6. Normalized (a) on and (b) off time probability distributions for the CdSeTe/ZnS quantum dots in PS host matrices at excitation pulse laser frequencies of 2, 2, and 500 MHz and cw excitation intensities of 120 (squares), 480 (circles), and 120 W/cm^2 (triangles), respectively. Solid lines show fitting to power-law decay.

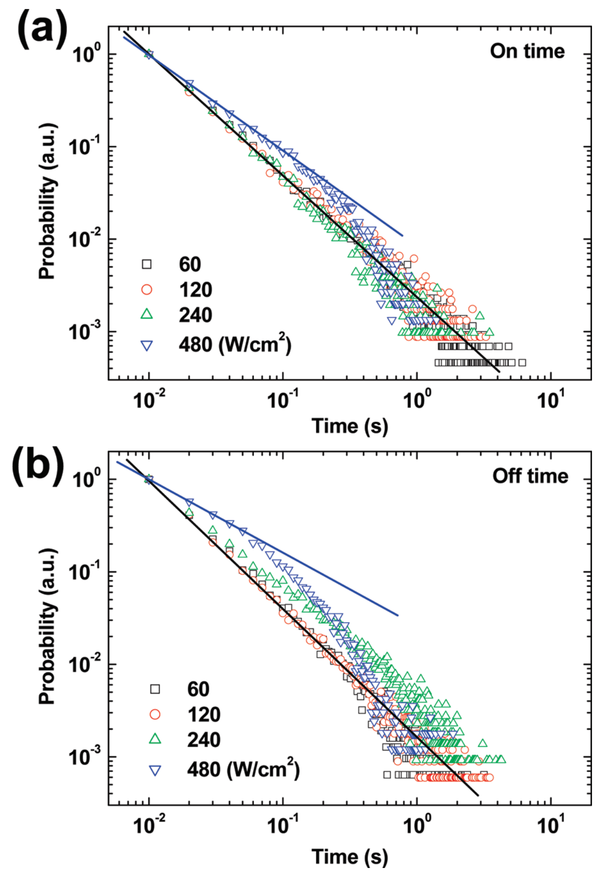

Figure 7. Normalized (a) on and (b) off time probability distributions for the CdSeTe/ZnS quantum dots in PS host matrices at excitation intensities of 60 (squares), 120 (circles), 240 (triangles), and 480 (inverted triangles) W/cm^2, respectively. Solid lines show fitting to power-law decay.

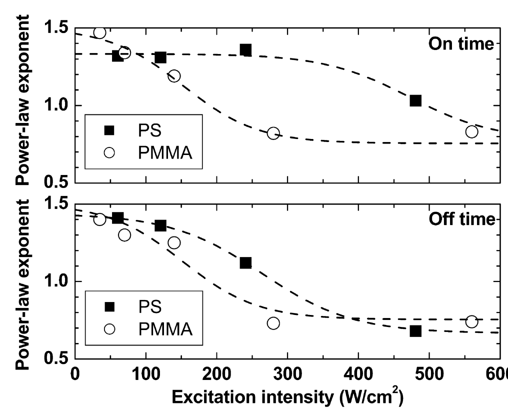

Figure 8. Excitation power dependences of power-law exponents of on (upper figure) and off (lower figure) states for the CdSeTe/ZnS quantum dots in PS (closed squares) and PMMA (open circles) matrices. Dashed lines show fitting to experimental data.

for the PMMA host matrices. Figure 8 shows the excitation intensity dependence of the power-law exponent for the CdSeTe/ZnS quantum dots in PS and PMMA host matrices. The power-law exponent slowly decreases with an increase in excitation intensity in contrast to the case for the PMMA host matrices.

Discussion

Origin of Different Cutoff Times in on and off States. For the CdSeTe/ZnS quantum dots, cutoff times were observed for both the on and off states. However, for the CdSe/ZnS quantum dots, the cutoff time was observed only for the on state. On the basis of the Tang−Marcus model, we discuss the origin of the different cutoff times for the on and off states. In the Tang−Marcus model, the cutoff time (τ_c) for on and off states is given by[14]

$$\tau_{c,k} = \frac{4\tau_L\langle z^2\rangle}{(z_c - x_{0,k})^2} \quad (k = \text{on or off}) \tag{14}$$

where z_c is the crossing point between the free energy potentials for the on and off states, and $x_{0,\text{on}} = 0$ and $x_{0,\text{off}} = 1$ are the lowest free energy potentials for the on and off state, respectively. From eq 8, since τ_L is a parameter determined by the media,[15,20] its value for the on and off states is the same. From eq 11, it is seen that $\langle z^2\rangle$ is dependent on the slope (κ) of the potential curve. κ is given by[19,20]

$$\kappa = 2e^2\left(\frac{1}{\varepsilon_\infty} - \frac{1}{\varepsilon_0}\right)\left(\frac{1}{2a_A} + \frac{1}{2a_B} - \frac{1}{r}\right) \tag{15}$$

where e is the elementary electric charge, a_A and a_B are the radii of the donor and acceptor sites, respectively, and r is the distance between the centers of the donor and acceptor sites. From eq 15, parameter κ has the same value for the neutral excited state and the charge-separated state. This means that $\langle z^2\rangle$ also shows the same value for the on and off states. Therefore, we expect that the difference in the cutoff times for the on and off states is dependent only on z_c. This suggests that z_c is significantly different for the CdSeTe/ZnS and CdSe/ZnS quantum dots. The crossing position of z_c is determined by the free energy gap (ΔG^0) between the potential lower levels for

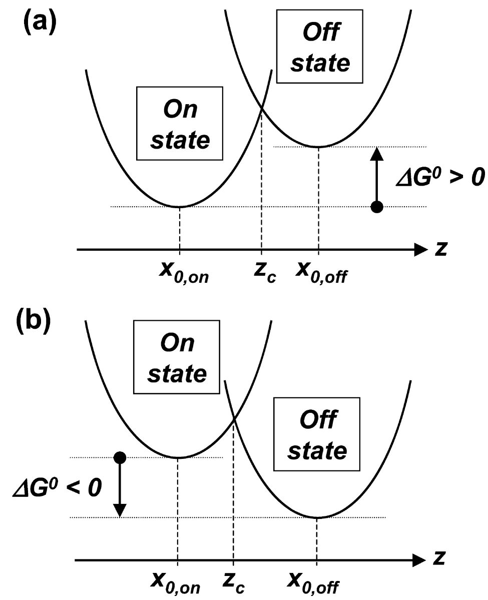

Figure 9. Free energy potentials in reaction coordinate space at the energy gap between the lower level of the on state and that of the off state (ΔG^0) for positive and negative values.

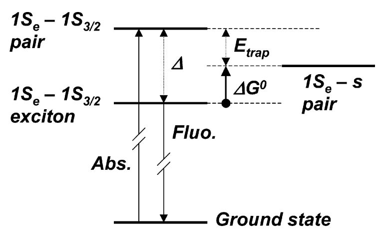

Figure 10. Energy diagram of NC quantum dots. $1S_{3/2}$, $1S_e$, and s are the highest valence level, lowest conduction level, and surface level, respectively. Δ and E_{trap} indicate the Stokes shift and energy gap between $1S_{3/2}$ and s, respectively.

the on and off states. Figure 9 shows schematic diagrams of the free energy potentials for the on and off states at (a) $\Delta G^0 > 0$ and (b) $\Delta G^0 < 0$. At $\Delta G^0 > 0$, since z_c is located near the potential lower level of $x_{0,off}$, the cutoff time for the on state is shorter than that for the off state. The CdSe/ZnS quantum dots correspond to this case. At $\Delta G^0 < 0$, since z_c is located near the potential lower level of $x_{0,on}$, the cutoff time for the on state is longer than for the off state. Subsequently, to understand the dependence of the cutoff time on the materials comparing the NC quantum dot, we discuss ΔG^0 in the two NC quantum dots. Figure 10 shows a schematic illustration of the electronic levels of NC quantum dots.[17,23−25] The energy of the neutral excited state is lower than the band gap between the conduction-band and valence-band edges due to a Stokes shift (Δ meV). The energy of the charge-separated state is equal to the energy gap between the conduction-band edge and the surface levels. Thus, ΔG^0 is given by

$$\Delta G^0 = \Delta - E_{trap} \tag{16}$$

where E_{trap} is the energy gap between the valence-band edge and the surface levels. For the CdSe/ZnS quantum dots, theoretical calculations show that E_{trap} is 50 meV;[17] a Δ of 70 meV was estimated experimentally from the energy gap between the lowest energy absorption peak and the fluorescence peak. Thus, for the CdSe/ZnS quantum dots, a ΔG^0 of 20 meV was obtained. This value correlates reasonably well with the

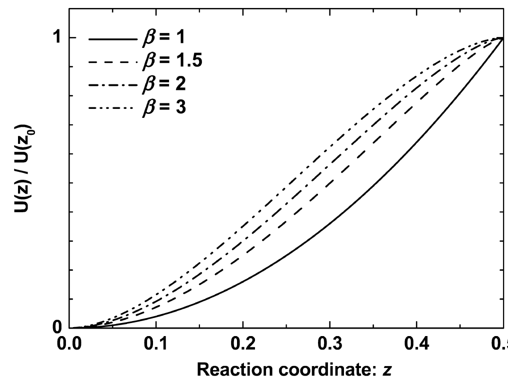

Figure 11. Normalized free energy potentials in reaction coordinate space, simulated by eq 6 at β of 1 (solid), 1.5 (dash), 2 (dash dot), and 3 (dash dot dot) respectively. In this simulation, $z_0 = 0.5$, $\tau = 1$ s, and $\phi_0 = 50$ meVs.

experimental results. For the CdSeTe/ZnS quantum dots, the calculation of ΔG^0 is difficult, since E_{trap} was not derived and the lowest absorption band was broad.[25] However, it is expected that ΔG^0 for the CdSeTe/ZnS quantum dots is nearly zero since the cutoff time for the on and off states is almost the same, as shown in Figure 4.

Excitation Intensity Dependence of the Power-Law Exponent. For the CdSeTe/ZnS quantum dots in PMMA matrices, the power-law exponent of the blinking statistics is significantly dependent on the excitation intensity. Here, we discuss the excitation intensity dependence of the power-law exponent. In the Tang-Marcus model, for Debye dielectric media the power-law exponent is -1.5.[14] For the Cole-Davidson dielectric media, the power-law exponent is $-2 + \beta_{CD}/2$.[15] β_{CD} is the Cole-Davidson parameter and generally ranges in value between zero and one.[27] Therefore, the model disagrees with the power-law exponent, which is below -1.5. However, based upon the Tang−Marcus model analogy for Cole-Davidson dielectric media,[15] $\langle z_n(t)\rangle$, which corresponds to a power-law exponent of $-2 + \beta/2$ ($\beta > 1$), is expected to be the following:

$$\langle z_n(t)\rangle = z_c \exp\left[-\left(\frac{t}{\tau}\right)^{\beta}\right] \quad (\beta > 1) \tag{17}$$

We also estimated $U(z)$, corresponding to eq 17, by using eq 6 with the overdamped limit. *Mathematica* was used for the simulations. Figure 11 shows the simulation results for $U(z)$ corresponding to several β values. According to linear response approximation between polarization and the electric field, β is 1 and the free energy forms a harmonic potential at the reaction coordinate.[19,20] With an increase in β, the potential bends in the surrounding area of the crossing position at $z_c = 0.5$. These anharmonic potentials are similar to the potential used in the model for optical nonlinearities.[27] Therefore, these results suggest a nonlinear interaction, which affects the potential at reaction coordinates. The candidate for the nonlinear interaction is an interaction between a strong electric field and polarizations in the dielectric media.[28,29] In the Tang−Marcus model, the electric field formed by carriers separated from the neutral excited state is independent of the excitation intensity. Thus, the nonlinear interaction between a strong electric field and polarizations in the dielectric media could result in an increase in the dielectric constant with the excitation intensity. For the independent power-law exponent of the off state for CdSe/ZnS quantum dots in PMMA matrices, the proposed model can be explained through the following mechanism. Since the crossing position of z_c is close to the potential lower level of $x_{0,off}$, the potential bending occurs at the reaction coordinate over z_c. As a result, the free energy potential for the off state maintains its harmonic potential with an increase in the excitation intensity.

We subsequently consider the correlation between the power-law exponent and the excitation intensity. Under cw laser excitation in the excitation intensity range of 175 to 700 W/cm^2, it was observed that the power-law exponent is independent of the excitation intensity.[11] The time-averaged excitation intensity of the cw laser is comparable to that of the pulse laser, while the transient peak intensity of the pulse laser is significantly higher than that of the cw laser. In addition, as shown in Figure 6, the change in the power-law exponent with an increase in the pulse laser frequency but same laser pulse energy is smaller than the change with an increase in the energy but same frequency. Therefore, these results suggest the strong correlation between the power-law exponent and the transient peak intensity.

In the framework of the proposed model, the measured strong correlation indicates that the dielectric constant of host matrices is increased transiently by the excitation pulses of the laser. Assuming that the transient time is much shorter than the time interval of the excitation pulses, the blinking statistics should be independent from the pulse laser frequency. As shown in Figure 6, however, the power-law exponent has a weak but discernible dependence on the frequency. Thus, the transient time is comparable to or longer than the excitation interval time of 500 ns.

Based upon these results, we propose possible mechanisms to explain the characteristic excitation intensity dependence on the dielectric constant. The first possible mechanism is that the increased dielectric constant is attributable to the electronically excited states of host matrices formed by two-photon excitation in media.[30,31] The lifetime of excited states of PMMA host matrices is greater than the time interval of the excitation pulses (500 ns) due to the n$-\pi^*$ transition (forbidden transition). Thus, it is possible for the increased dielectric constant to be affected in the time scale of blinking events. We also observed the blinking statistics for CdSeTe/ZnS quantum dots in PS media, as shown Figure 7. The experimental results shown in Figure 8 might be explained by the first possible mechanism as follows. The lifetime of the excited state in PS matrices is shorter than in PMMA matrices since this transition corresponds to $\pi-\pi^*$. The lifetime of the excited state in comparable to the time scale of the excitation time interval due to the formation of an excimer.[32] In the first possible model, the excited states in PS matrices weakly affect blinking events since the lifetime of PS is shorter than that of PMMA. In practice, the excitation intensity dependence of the power-law exponents in PS matrices is weaker than that in PMMA matrices (see Figure 8).

The second possible mechanism is laser-induced molecular reorientation. The polar molecular moieties in PMMA would align with the laser irradiation. The relaxation times of the molecular alignment would be longer than the laser excitation interval time. Thus, it is possible for the increased dielectric constant to be affected in the time scale of blinking events. The experimental results shown in Figure 8 might be explained by the second possible mechanism as follows. PMMA include the polar molecular moieties in contrast to PS. Therefore the degree of the laser-induced molecular reorientation in PS is smaller than in PMMA.

Conclusions

In this paper, we observed the blinking statistics for CdSeTe/ZnS and CdSe/ZnS quantum dots in polymer matrices under pulse excitation with peak intensities higher than that obtained with a cw laser; we were able to obtain new information on the blinking statistics. First, in the blinking statistics for CdSeTe/ZnS quantum dots, we observed the cutoff time for the off state as predicted by the Tang−Marcus model, the first time that this has been done. In addition, in the blinking statistics for both CdSeTe/ZnS and CdSe/ZnS quantum dots, we observed the excitation intensity dependence of the power-law exponent; this is an important new piece of information when considering possible mechanisms for fluorescence blinking. Using these new results, we investigated the validity of the Tang−Marcus model. To explain the excitation intensity dependence of the power-law exponent in the framework of the Tang−Marcus model, we proposed an extension of the Tang−Marcus model based on a diffusion-controlled model using anharmonic potentials. As a mechanism for anharmonic potential formation at the reaction coordinates, we suggested the interaction between a strong electric field and polarizations in the dielectric media. In addition, to explain the nonlinear interaction, we investigated pulsed laser frequency and host matrix dependences of the blinking statistics. We proposed several possible mechanisms to explain experimental results. The new obtained results provide significant insight into fluorescent blinking statistics in the framework of the Tang−Marcus model.

References and Notes

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